Multivariate Data Analysis Problem Bank: Solutions

Numerical answers are rounded to about three decimals.

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Part I. Distributions and Inference

Solution 1 [Ch. 1 — centering matrix and the rank of $\mathbf{S}$]

(a) $\mathbf{H}_n^\top = \mathbf{I}_n - \frac{1}{n}(\mathbf{1}\mathbf{1}^\top)^\top = \mathbf{H}_n$. Since $\mathbf{1}_n^\top \mathbf{1}_n = n$, $\mathbf{H}_n^2 = \mathbf{I}_n - \frac{2}{n}\mathbf{1}\mathbf{1}^\top + \frac{1}{n^2}\mathbf{1}(\mathbf{1}^\top\mathbf{1})\mathbf{1}^\top = \mathbf{H}_n$ and $\mathbf{H}_n\mathbf{1}_n = \mathbf{1}_n - \frac{1}{n}\mathbf{1}_n \cdot n = \mathbf{0}$. Being symmetric idempotent, its eigenvalues are 0 or 1 and $\mathrm{rank}\mathbf{H}_n = \mathrm{tr}\mathbf{H}_n = n - \frac{1}{n}\mathrm{tr}(\mathbf{1}\mathbf{1}^\top) = n - 1$.

(b) The ith row of $\mathbf{H}_n\mathbf{X}$ is $\mathbf{x}_i^\top - \frac{1}{n}\sum_k \mathbf{x}_k^\top = (\mathbf{x}_i - \bar{\mathbf{x}})^\top$, so $\mathbf{X}^\top\mathbf{H}_n\mathbf{X} = (\mathbf{H}_n\mathbf{X})^\top(\mathbf{H}_n\mathbf{X}) = \sum_i(\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top = (n-1)\mathbf{S}$. Then

$$\mathrm{rank}\mathbf{S} = \mathrm{rank}\{(\mathbf{H}_n\mathbf{X})^\top(\mathbf{H}_n\mathbf{X})\} = \mathrm{rank}(\mathbf{H}_n\mathbf{X}) \le \min\{\mathrm{rank}\mathbf{H}_n, \mathrm{rank}\mathbf{X}\} \le \min(n-1, p).$$

(c) If $p \ge n$, $\mathrm{rank}\mathbf{S} \le n - 1 < p$, so the $p \times p$ matrix $\mathbf{S}$ is singular. Then $\mathbf{S}^{-1/2}$ does not exist (no sphering) and $(\mathbf{x} - \bar{\mathbf{x}})^\top\mathbf{S}^{-1}(\mathbf{x} - \bar{\mathbf{x}})$ is undefined. Geometrically the n points lie in an affine subspace of dimension at most $n-1$, so some directions have zero estimated variance. Alternatives: diagonal (naive Bayes) or regularized ($\mathbf{S} + \lambda\mathbf{I}$) versions.

Solution 2 [Ch. 1 — geometry of a small data set]

(a) $\bar{\mathbf{x}} = (5, 8)^\top$; deviations $(-2,-1), (0,1), (-1,-4), (3,4)$; $s_{11} = 14/3$, $s_{22} = 34/3$, $s_{12} = 18/3 = 6$:

$$\mathbf{S} = \begin{pmatrix} 14/3 & 6 \\ 6 & 34/3 \end{pmatrix} \approx \begin{pmatrix} 4.667 & 6 \\ 6 & 11.333 \end{pmatrix}, \qquad r_{12} = \frac{6}{\sqrt{(14/3)(34/3)}} = 0.825.$$

(b) $|\mathbf{S}| = 476/9 - 36 = 152/9$ and $\lambda = 8 \pm \sqrt{64 - 152/9} = 8 \pm 6.864$: $\lambda_1 = 14.864$, $\lambda_2 = 1.136$, with $\mathbf{u}_1 \approx (0.507, 0.862)^\top$, $\mathbf{u}_2 \approx (-0.862, 0.507)^\top$. The major axis is along $\mathbf{u}_1$ (about 59.5°) with half-length $\sqrt{c\lambda_1} = 3.855\sqrt{c}$; the minor half-length is $\sqrt{c\lambda_2} = 1.066\sqrt{c}$.

(c) With $\mathbf{S}^{-1} = \frac{9}{152}\begin{pmatrix} 34/3 & -6 \\ -6 & 14/3 \end{pmatrix}$: $d_i^2 = 1.539,\ 0.276,\ 2.250,\ 1.934$, total 6.000. In general

$$\sum_i d_i^2 = \sum_i \operatorname{tr}\{\mathbf{S}^{-1}(\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top\} = \operatorname{tr}\Big\{\mathbf{S}^{-1}\sum_i (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top\Big\} = \operatorname{tr}\{(n-1)\mathbf{I}_p\} = (n-1)p,$$

here $3 \cdot 2 = 6$.

(d) $\mathbf{S}^{-1/2} = \mathbf{U}\boldsymbol{\Lambda}^{-1/2}\mathbf{U}^\top \approx \begin{pmatrix} 0.764 & -0.297 \\ -0.297 & 0.434 \end{pmatrix}$ (the sphered data have covariance $\mathbf{I}_2$).

Solution 3 [Ch. 1 — rank deficiency]

(a) $\mathbf{a} = (1, 1, -1)^\top$: $\mathbf{a}^\top \mathbf{x}_i = x_{i1} + x_{i2} - x_{i3} = 0$, so $\mathbf{a}^\top \bar{\mathbf{x}} = 0$ and $\mathbf{a}^\top (\mathbf{x}_i - \bar{\mathbf{x}}) = 0$. Hence $\mathbf{S}\mathbf{a} = \frac{1}{n-1} \sum_i (\mathbf{x}_i - \bar{\mathbf{x}})\{(\mathbf{x}_i - \bar{\mathbf{x}})^\top \mathbf{a}\} = \mathbf{0}$.

(b) $\mathbf{d}_3 = \mathbf{d}_1 + \mathbf{d}_2$, so $\text{span}\{\mathbf{d}_1, \mathbf{d}_2, \mathbf{d}_3\} = \text{span}\{\mathbf{d}_1, \mathbf{d}_2\}$ has dimension ≤ 2 and $\text{rank}\mathbf{S} = \text{rank}\mathbf{X}_C \leq 2$.

(c) $\mathbf{S}\mathbf{a} = 0 \cdot \mathbf{a}$, so 0 is the smallest eigenvalue: the third PC has variance 0, direction $(1, 1, -1)^\top / \sqrt{3}$ (assuming the other two eigenvalues are positive).

Solution 4 [2024 Midterm Q1 — data matrix and a singular covariance]

(a) $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_{100}]^\top$ is 100×2: 100 rows (observations) and 2 columns (variables).

(b) $\bar{\mathbf{x}} = \frac{1}{n}\mathbf{X}^\top \mathbf{1}_n$ and $\mathbf{S} = \frac{1}{n-1}\mathbf{X}^\top(\mathbf{I}_n - \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top)\mathbf{X} = \frac{1}{n-1}\mathbf{X}^\top\mathbf{H}_n\mathbf{X}$ with $n = 100$.

(c) $|\mathbf{\Sigma}| = 36 - 36 = 0$, and in fact $\mathbf{\Sigma} = \mathbf{v}\mathbf{v}^\top$ with $\mathbf{v} = (2, -3)^\top$. Let $\mathbf{a} = (3, 2)^\top$; then $\mathbf{\Sigma}\mathbf{a} = (12 - 12, -18 + 18)^\top = \mathbf{0}$, so $\mathrm{Var}(\mathbf{a}^\top\mathbf{x}_i) = \mathbf{a}^\top\mathbf{\Sigma}\mathbf{a} = 0$ and $\mathbf{a}^\top\mathbf{x}_i = \mathbf{a}^\top\boldsymbol{\mu} = 3$ with probability one. Hence $\mathbf{a}^\top\bar{\mathbf{x}} = 3$ and $\mathbf{a}^\top(\mathbf{x}_i - \bar{\mathbf{x}}) = 0$ for all i, so

$$\mathbf{S}\mathbf{a} = \tfrac{1}{n-1}\sum_i (\mathbf{x}_i - \bar{\mathbf{x}})\{(\mathbf{x}_i - \bar{\mathbf{x}})^\top\mathbf{a}\} = \mathbf{0} \quad \Rightarrow \quad \mathrm{rank}(\mathbf{S}) \le 1.$$

On the other hand $\mathbf{w} = \mathbf{v}/\sqrt{13}$ gives $\mathrm{Var}(\mathbf{w}^\top\mathbf{x}_i) = \mathbf{w}^\top\mathbf{\Sigma}\mathbf{w} = 13 > 0$; since the $\mathbf{w}^\top\mathbf{x}_i$ are non-degenerate and $n = 100 \ge 2$, they are not all equal (a.s. for continuous, e.g. normal, data), so $\mathbf{S} \neq \mathbf{0}$. Therefore $\mathrm{rank}(\mathbf{S}) = 1$ (a.s.). **Spread**: all observations lie on the line $3x_1 + 2x_2 = 3$ through $\boldsymbol{\mu} = (1, 0)$, in direction $(2, -3)^\top$. The population correlation is $-6/\sqrt{4 \cdot 9} = -1$ (perfect negative linear relation); the total variance $\mathrm{tr}\mathbf{\Sigma} = 13$ is entirely along $\mathbf{v}$ (eigenvalues $13, 0$), and there is no variation perpendicular to the line. The data are effectively one-dimensional.

(d) $\mathbf{\Sigma}$ is singular, so $\mathbf{\Sigma}^{-1}$ and hence $(\mathbf{y} - \boldsymbol{\mu})^\top\mathbf{\Sigma}^{-1}(\mathbf{y} - \boldsymbol{\mu})$ do not exist. Moreover $\mathbf{y} - \boldsymbol{\mu} = (0, 1)^\top$ and $\mathbf{a}^\top(\mathbf{y} - \boldsymbol{\mu}) = 2 \neq 0$: $\mathbf{y}$ is off the support line, a point that cannot occur under this distribution, so its statistical distance is infinite (sphering would have to divide by the zero variance in direction $\mathbf{a}$). (For points on the line, $\mathbf{y} = \boldsymbol{\mu} + t\mathbf{v}$, one can use the generalized inverse $\mathbf{\Sigma}^+ = \mathbf{v}\mathbf{v}^\top/169$ and get $d^2 = t^2\,\mathbf{v}^\top\mathbf{\Sigma}^+\mathbf{v} = t^2$: the point is $t\sqrt{13}$ away along a direction with SD $\sqrt{13}$.)

(e) $\bar{\mathbf{x}} \sim N_2(\boldsymbol{\mu}, \frac{1}{100}\mathbf{\Sigma}) = N_2\Big(\begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}0.04 & -0.06\\ -0.06 & 0.09\end{pmatrix}\Big)$, a singular normal supported on the same line $(3\bar{x}_1 + 2\bar{x}_2 = 3$ a.s.); it is defined via the general (projection) definition since no density exists.

Solution 5 [Ch. 1, 3 — singular population covariance]

(a) $\mathrm{Var}(\mathbf{a}^\top \mathbf{X}) = \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a} = 0$ and $\mathrm{E}(\mathbf{a}^\top \mathbf{X}) = \mathbf{a}^\top \boldsymbol{\mu}$, so by Chebyshev $P(|\mathbf{a}^\top \mathbf{X} - \mathbf{a}^\top \boldsymbol{\mu}| > \epsilon) = 0$ for every $\epsilon > 0$; hence $\mathbf{a}^\top \mathbf{X} = \mathbf{a}^\top \boldsymbol{\mu}$ a.s. ($\mathbf{X}$ lies on the hyperplane $\{\mathbf{x} : \mathbf{a}^\top (\mathbf{x} - \boldsymbol{\mu}) = 0\}$.)

(b) By (a), $\mathbf{a}^\top \mathbf{x}_i = \mathbf{a}^\top \boldsymbol{\mu}$ for all i a.s., so $\mathbf{a}^\top \bar{\mathbf{x}} = \mathbf{a}^\top \boldsymbol{\mu}$ and $\mathbf{a}^\top (\mathbf{x}_i - \bar{\mathbf{x}}) = 0$; thus $\mathbf{S}\mathbf{a} = \mathbf{0}$. Applying this to every $\mathbf{a}$ in the null space of $\boldsymbol{\Sigma}$: $\mathrm{null}(\boldsymbol{\Sigma}) \subseteq \mathrm{null}(\mathbf{S})$, hence $\mathrm{rank}\mathbf{S} = p - \dim \mathrm{null}(\mathbf{S}) \leq p - \dim \mathrm{null}(\boldsymbol{\Sigma}) = \mathrm{rank}\boldsymbol{\Sigma}$.

(c) $\boldsymbol{\Sigma}$ is not invertible, so $\boldsymbol{\Sigma}^{-1}$ does not exist. Points with $\mathbf{a}^\top (\mathbf{y} - \boldsymbol{\mu}) \neq 0$ are outside the support (probability zero); their "distance" is infinite since sphering would divide by a zero standard deviation in direction $\mathbf{a}$. Problem 4(d) is an instance.

Solution 6 [2024 Midterm Q2 — extremes of a quadratic form]

$4x_1^2 + 4x_2^2 + 6x_1x_2 = \mathbf{x}^\top \mathbf{A}\mathbf{x}$ with $\mathbf{A} = \begin{pmatrix} 4 & 3 \\ 3 & 4 \end{pmatrix}$, eigenvalues 7 (eigenvector $(1,1)^\top/\sqrt{2}$) and 1 (eigenvector $(1,-1)^\top/\sqrt{2}$). With $\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$ and $\mathbf{z} = \mathbf{U}^\top\mathbf{x}$ ($\|\mathbf{z}\| = 1$), $\mathbf{x}^\top\mathbf{A}\mathbf{x} = 7z_1^2 + z_2^2 \in [1, 7]$. Hence the maximum is 7 at $\mathbf{x} = \pm(1,1)^\top/\sqrt{2}$ and the minimum is 1 at $\mathbf{x} = \pm(1,-1)^\top/\sqrt{2}$. (Check: $4 \cdot \frac{1}{2} + 4 \cdot \frac{1}{2} + 6 \cdot \frac{1}{2} = 7$.)

Solution 7 [Ch. 2 — an elementary inequality and a third MLE proof]

(a) With eigenvalues $a_i > 0$ of $\mathbf{A}$, $\mathrm{tr}\mathbf{A} - \log|\mathbf{A}| - p = \sum_i (a_i - \log a_i - 1) \geq 0$ by Theorem 2.14, with equality iff all $a_i = 1$, i.e. $\mathbf{A} = \mathbf{I}_p$ (symmetric).

(b) $\mathbf{B} = \mathbf{S}_0^{1/2}\boldsymbol{\Sigma}^{-1}\mathbf{S}_0^{1/2} \succ 0$, $\log|\mathbf{B}| = \log|\mathbf{S}_0| - \log|\boldsymbol{\Sigma}|$, $\mathrm{tr}\mathbf{B} = \mathrm{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{S}_0)$. So $f(\boldsymbol{\Sigma}) = \log|\mathbf{S}_0| - \log|\mathbf{B}| + \mathrm{tr}\mathbf{B}$ and $f(\mathbf{S}_0) = \log|\mathbf{S}_0| + p$, giving $f(\boldsymbol{\Sigma}) - f(\mathbf{S}_0) = \mathrm{tr}\mathbf{B} - \log|\mathbf{B}| - p \geq 0$, with equality iff $\mathbf{B} = \mathbf{I} \iff \boldsymbol{\Sigma} = \mathbf{S}_0$. The unique minimizer is $\mathbf{S}_0$ — the shortest proof, using neither von Neumann nor calculus.

Solution 8 [Ch. 2 — block-matrix formulas]

(a) $(\mathbf{u}\mathbf{v}^\top)\mathbf{u} = (\mathbf{v}^\top\mathbf{u})\mathbf{u}$, so $\mathbf{v}^\top\mathbf{u}$ is an eigenvalue of the rank-one $\mathbf{u}\mathbf{v}^\top$ and the other $p-1$ eigenvalues are 0. Thus $\mathbf{I}+\mathbf{u}\mathbf{v}^\top$ has eigenvalues $1+\mathbf{v}^\top\mathbf{u}$ and 1 ($p-1$ times) and determinant $1+\mathbf{v}^\top\mathbf{u}$. Then $|\mathbf{A}+\mathbf{u}\mathbf{v}^\top| = |\mathbf{A}||\mathbf{I}+\mathbf{A}^{-1}\mathbf{u}\mathbf{v}^\top| = |\mathbf{A}|(1+\mathbf{v}^\top\mathbf{A}^{-1}\mathbf{u})$.

(b) With $\mathbf{L} = \begin{pmatrix} \mathbf{I} & -\mathbf{A}_{12}\mathbf{A}_{22}^{-1} \\ \mathbf{0} & \mathbf{I} \end{pmatrix}$, $\mathbf{L}\mathbf{A}\mathbf{L}^\top = \operatorname{diag}(\mathbf{A}_{11\cdot 2}, \mathbf{A}_{22})$. Hence $\mathbf{A}^{-1} = \mathbf{L}^\top \operatorname{diag}(\mathbf{A}_{11\cdot 2}^{-1}, \mathbf{A}_{22}^{-1})\mathbf{L}$. The first block row of $\mathbf{L}^\top$ is $(\mathbf{I}, \mathbf{0})$ and the first block column of $\mathbf{L}$ is $(\mathbf{I}, \mathbf{0})^\top$, so the $(1,1)$ block is $\mathbf{A}_{11\cdot 2}^{-1}$.

(c) By (a), $|\mathbf{W}+\mathbf{b}\mathbf{b}^\top| = |\mathbf{W}|(1+\mathbf{b}^\top\mathbf{W}^{-1}\mathbf{b})$. This single identity gives both $\Lambda^{2/n} = (1+T^2/(n-1))^{-1}$ and, for $g=2$, $\Lambda^* = (1+T^2/(n-2))^{-1}$.

Solution 9 [Ch. 3 — linear transformations and conditional distributions]

(a) $\boldsymbol{\Sigma}_{12} = (1, 0)$, $\boldsymbol{\Sigma}_{22} = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}$, $\boldsymbol{\Sigma}_{22}^{-1} = \frac{1}{5}\begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}$, $\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1} = (0.4, -0.2)$, $\boldsymbol{\Sigma}_{11\cdot 2} = 4 - 0.4 = 3.6$:

$$X_1 \mid (x_2, x_3) \sim N\big(1 + 0.4(x_2 - 2) - 0.2x_3,\ 3.6\big).$$

(b) $\mathbf{A} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & -1 \end{pmatrix}$, $\mathbf{A}\boldsymbol{\mu} = (3, 1)^\top$, $\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top = \begin{pmatrix} 9 & 4 \\ 4 & 6 \end{pmatrix}$: $N_2\Big((3, 1)^\top, \begin{pmatrix} 9 & 4 \\ 4 & 6 \end{pmatrix}\Big)$.

(c) $\mathrm{Cov}(X_1, X_3) = 0$ and the pair is jointly normal: independent. $\mathrm{Cov}((X_1, X_2), X_3) = (0, 1)^\top \neq \mathbf{0}$: not independent. $\mathrm{Cov}(X_1, X_2 - \frac{1}{4}X_1) = 1 - \frac{1}{4} \cdot 4 = 0$ with joint normality: independent (this is the BLP residual $X_2 - \mathrm{BLP}(X_2 \mid X_1)$).

Solution 10 [Ch. 3 — BLP, multiple and partial correlation]

(a) $\boldsymbol{\Sigma}_{XX} = \begin{pmatrix} 2 & 1 \\ 1 & 4 \end{pmatrix}$, $\boldsymbol{\Sigma}_{XX}^{-1} = \frac{1}{7}\begin{pmatrix} 4 & -1 \\ -1 & 2 \end{pmatrix}$, $\boldsymbol{\Sigma}_{XY} = (3,4)^\top$, $\boldsymbol{\beta} = \boldsymbol{\Sigma}_{XX}^{-1}\boldsymbol{\Sigma}_{XY} = (8/7, 5/7)^\top$:

$$\mathrm{BLP}(Y \mid \mathbf{X}) = 5 + \tfrac{8}{7}(X_1 - 1) + \tfrac{5}{7}(X_2 - 2) = \tfrac{17}{7} + \tfrac{8}{7}X_1 + \tfrac{5}{7}X_2.$$

Residual variance $\sigma_Y^2 - \boldsymbol{\Sigma}_{YX}\boldsymbol{\beta} = 10 - 44/7 = 26/7 \approx 3.714$.

(b) $\rho^2_{\mathrm{mult}} = \frac{44/7}{10} = 0.6286$, $\rho_{\mathrm{mult}} = 0.793$.

(c) $\sigma_{YX_1 \cdot X_2} = 3 - \frac{4 \cdot 1}{4} = 2$, $\mathrm{Var}(\tilde{Y}) = 10 - \frac{16}{4} = 6$, $\mathrm{Var}(\tilde{X}_1) = 2 - \frac{1}{4} = 1.75$; partial correlation $2/\sqrt{6 \cdot 1.75} = 0.617$, slightly below the marginal $3/\sqrt{20} = 0.671$ (part of the association runs through X_2).

(d) W.l.o.g. standardize. $\tilde{Y} = Y - \rho_{YZ}Z$, $\tilde{X} = X - \rho_{XZ}Z$, so $\mathrm{Cov}(\tilde{Y}, \tilde{X}) = \rho_{YX} - \rho_{YZ}\rho_{XZ}$, $\mathrm{Var}\tilde{Y} = 1 - \rho^2_{YZ}$, $\mathrm{Var}\tilde{X} = 1 - \rho^2_{XZ}$. Numerically $(0.6708 - 0.6325 \cdot 0.3536)/\sqrt{0.6 \cdot 0.875} = 0.617$.

Solution 11 [2024 Midterm Q4 — a recursive model]

(a) $\mathrm{Var}(X_1) = \sigma^2$, $\mathrm{Var}(X_2) = \beta_{12}^2\sigma^2 + \sigma^2$, $\mathrm{Cov}(X_1, X_2) = \beta_{12}\sigma^2$. (X_1, X_2) is jointly normal (linear in independent normals), so by Theorem 3.11

$$\mathrm{Var}(X_1 \mid X_2) = \sigma^2 - \frac{\beta_{12}^2\sigma^4}{(1+\beta_{12}^2)\sigma^2} = \frac{\sigma^2}{1+\beta_{12}^2}.$$

(b) Write the model as $\boldsymbol{\varepsilon} = (\mathbf{I} - \mathbf{L})\mathbf{X}$ with $\mathbf{L} = \begin{pmatrix} 0 & 0 & 0 \\ \beta_{12} & 0 & 0 \\ \beta_{13} & \beta_{23} & 0 \end{pmatrix}$. $\mathbf{I} - \mathbf{L}$ is unit lower triangular (invertible, determinant 1) and $\mathrm{Cov}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}$, so $\boldsymbol{\Sigma} = \sigma^2(\mathbf{I} - \mathbf{L})^{-1}(\mathbf{I} - \mathbf{L})^{-\top}$ and

$$\boldsymbol{\Sigma}^{-1} = \frac{1}{\sigma^2}(\mathbf{I} - \mathbf{L})^\top(\mathbf{I} - \mathbf{L}) = \frac{1}{\sigma^2}\begin{pmatrix} 1+\beta_{12}^2+\beta_{13}^2 & -\beta_{12}+\beta_{13}\beta_{23} & -\beta_{13} \\ -\beta_{12}+\beta_{13}\beta_{23} & 1+\beta_{23}^2 & -\beta_{23} \\ -\beta_{13} & -\beta_{23} & 1 \end{pmatrix}.$$

(Entries are inner products of the columns $(1, -\beta_{12}, -\beta_{13})$, $(0, 1, -\beta_{23})$, $(0, 0, 1)$ of $\mathbf{I}-\mathbf{L}$.) By Problem 12, $\omega_{13} = -\beta_{13}/\sigma^2 = 0 \iff X_1 \perp\!\!\!\perp X_3 \mid X_2$, and $(\boldsymbol{\Sigma}^{-1})_{33} = 1/\sigma^2$ gives $\mathrm{Var}(X_3 \mid X_1, X_2) = \sigma^2$, as the model says.

Solution 12 [Ch. 3 — the precision matrix]

(a) Order $\mathbf{X} = (X_i, \mathbf{X}_{-i}^\top)^\top$ and partition $\boldsymbol{\Sigma}$ and $\boldsymbol{\Omega}$ conformably. By the block-inverse formula (Ch. 2),

$$\omega_{ii} = (\boldsymbol{\Sigma}^{-1})_{11} = \left(\sigma_{ii} - \boldsymbol{\Sigma}_{i,-i}\boldsymbol{\Sigma}_{-i,-i}^{-1}\boldsymbol{\Sigma}_{-i,i}\right)^{-1} = \frac{1}{\operatorname{Var}(X_i \mid \mathbf{X}_{-i})}$$

by Theorem 3.11. Also $\boldsymbol{\Omega}_{i,-i} = -\omega_{ii}\boldsymbol{\Sigma}_{i,-i}\boldsymbol{\Sigma}_{-i,-i}^{-1}$ (off-diagonal block of the inverse), so the regression coefficient is $\boldsymbol{\Sigma}_{i,-i}\boldsymbol{\Sigma}_{-i,-i}^{-1} = -\boldsymbol{\Omega}_{i,-i}/\omega_{ii}$ and $\mathrm{E}(X_i \mid \mathbf{x}_{-i}) = \mu_i - \frac{1}{\omega_{ii}}\sum_{j\neq i}\omega_{ij}(x_j - \mu_j)$.

(b) Let $a = \{i, j\}$, b the rest. The partial covariance of $\mathbf{X}_a$ given $\mathbf{X}_b$ is the Schur complement $\boldsymbol{\Sigma}_{aa\cdot b}$, and by the block-inverse formula $\boldsymbol{\Sigma}_{aa\cdot b} = (\boldsymbol{\Omega}_{aa})^{-1}$. With $\boldsymbol{\Omega}_{aa} = \begin{pmatrix} \omega_{ii} & \omega_{ij} \\ \omega_{ij} & \omega_{jj} \end{pmatrix}$,

$$\boldsymbol{\Sigma}_{aa\cdot b} = \frac{1}{\omega_{ii}\omega_{jj} - \omega_{ij}^2}\begin{pmatrix} \omega_{jj} & -\omega_{ij} \\ -\omega_{ij} & \omega_{ii} \end{pmatrix} \;\Rightarrow\; \operatorname{pcorr}(X_i, X_j; \text{rest}) = \frac{-\omega_{ij}}{\sqrt{\omega_{ii}\omega_{jj}}}.$$

Under normality the partial covariance is the conditional covariance, so $X_i \perp\!\!\!\perp X_j \mid \text{rest} \iff \omega_{ij} = 0$.

(c) In Problem 11, $\omega_{13} = -\beta_{13}/\sigma^2$ and $\omega_{23} = -\beta_{23}/\sigma^2$: $X_1 \perp\!\!\!\perp X_3 \mid X_2$ iff $\beta_{13} = 0$ (no direct arrow $X_1 \to X_3$). $\omega_{12} = (-\beta_{12} + \beta_{13}\beta_{23})/\sigma^2$ can vanish even when $\beta_{12} \neq 0$, because conditioning on the common child X_3 induces dependence between X_1 and X_2.

Solution 13 [Ch. 3 — uncorrelated normal marginals that are not independent]

(a) $P(X_2 \le x) = \frac{1}{2}P(X_1 \le x) + \frac{1}{2}P(-X_1 \le x) = \Phi(x)$ by symmetry; $\mathrm{Cov}(X_1, X_2) = \mathrm{E}(WX_1^2) = \mathrm{E}W\,\mathrm{E}X_1^2 = 0$.

(b) $|X_2| = |X_1|$, so e.g. $P(|X_1| > 1, |X_2| < 1) = 0 \neq P(|X_1| > 1)P(|X_2| < 1) > 0$.

(c) $X_1 + X_2 = (1 + W)X_1$ equals 0 with probability $\frac{1}{2}$ and $2X_1$ otherwise; $P(X_1 + X_2 = 0) = \frac{1}{2}$, so it is neither normal nor a point mass. A non-normal linear combination means (X_1, X_2) is not jointly normal; Theorem 3.9(b) assumes joint normality, so there is no contradiction.

Solution 14 [Ch. 3 — quadratic forms]

(a) $\mathbf{H}_n\mathbf{1} = \mathbf{0}$ gives $\mathbf{Y}^\top\mathbf{H}_n\mathbf{Y} = \mathbf{Z}^\top\mathbf{H}_n\mathbf{Z}$ with $\mathbf{Z} = \mathbf{Y} - \mu\mathbf{1} \sim N_n(\mathbf{0}, \sigma^2\mathbf{I})$. $\mathbf{H}_n$ is symmetric idempotent of rank $n-1$, so Theorem 3.20 gives χ^2_{n-1}. $\bar{Y} - \mu = \frac{1}{n}\mathbf{1}^\top\mathbf{Z} = \mathbf{B}\mathbf{Z}$ with $\mathbf{B}\mathbf{H}_n = \frac{1}{n}(\mathbf{H}_n\mathbf{1})^\top = \mathbf{0}$, so by Theorem 3.23 (applied to $\mathbf{Z}/\sigma$) $\bar{Y} \perp\!\!\!\perp S^2$.

(b) $X_1^2 + X_2^2 = \mathbf{X}^\top\mathbf{I}\mathbf{X}$; by Theorem 3.21 it is chi-square iff $\boldsymbol{\Sigma}$ is idempotent. $\boldsymbol{\Sigma}^2 = \begin{pmatrix} 1+\rho^2 & 2\rho \\ 2\rho & 1+\rho^2 \end{pmatrix} = \boldsymbol{\Sigma} \iff \rho = 0$ (then χ^2_2). The second form is $\mathbf{X}^\top\boldsymbol{\Sigma}^{-1}\mathbf{X}$ ($\boldsymbol{\Sigma}^{-1} = \frac{1}{1-\rho^2}\begin{pmatrix} 1 & -\rho \\ -\rho & 1 \end{pmatrix}$), which is χ^2_2 for all $|\rho| < 1$.

Solution 15 [2024 Midterm Q5 — proofs]

(a) Since $\sum_i(\mathbf{X}_i - \bar{\mathbf{X}}) = \mathbf{0}$,

$$\sum_i(\mathbf{X}_i - \bar{\mathbf{X}})(\mathbf{X}_i - \bar{\mathbf{X}})^\top = \sum_i(\mathbf{X}_i - \boldsymbol{\mu})(\mathbf{X}_i - \boldsymbol{\mu})^\top - n(\bar{\mathbf{X}} - \boldsymbol{\mu})(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top.$$

$\mathrm{E}(\mathbf{X}_i - \boldsymbol{\mu})(\mathbf{X}_i - \boldsymbol{\mu})^\top = \boldsymbol{\Sigma}$, and $\mathrm{Cov}(\bar{\mathbf{X}}) = \frac{1}{n^2}\sum_j\sum_\ell \mathrm{E}\{(\mathbf{X}_j - \boldsymbol{\mu})(\mathbf{X}_\ell - \boldsymbol{\mu})^\top\} = \frac{1}{n^2}n\boldsymbol{\Sigma} = \frac{1}{n}\boldsymbol{\Sigma}$ (cross terms vanish by independence). Hence $\mathrm{E}\sum_i(\mathbf{X}_i - \bar{\mathbf{X}})(\mathbf{X}_i - \bar{\mathbf{X}})^\top = n\boldsymbol{\Sigma} - \boldsymbol{\Sigma} = (n-1)\boldsymbol{\Sigma}$ and $\mathrm{E}(\mathbf{S}_n) = \frac{n-1}{n}\boldsymbol{\Sigma} \neq \boldsymbol{\Sigma}$: biased.

(b) ($\Leftarrow$) $\mathbf{A} = \boldsymbol{\Gamma}\boldsymbol{\Lambda}\boldsymbol{\Gamma}^\top$ with $\boldsymbol{\Lambda} = \mathrm{diag}(1,\ldots,1,0,\ldots,0)$ (m ones, since eigenvalues are $0/1$ and there are $\mathrm{rank}\mathbf{A} = m$ nonzero ones). Let $\mathbf{z} = \frac{1}{\sigma}\boldsymbol{\Gamma}^\top\mathbf{X} \sim N_p(\mathbf{0}, \boldsymbol{\Gamma}^\top\boldsymbol{\Gamma}) = N_p(\mathbf{0}, \mathbf{I}_p)$. Then $Y = \mathbf{z}^\top\boldsymbol{\Lambda}\mathbf{z} = z_1^2 + \cdots + z_m^2 \sim \chi^2(m)$. ($\Rightarrow$) For symmetric $\mathbf{A} = \boldsymbol{\Gamma}\boldsymbol{\Lambda}\boldsymbol{\Gamma}^\top$ (any real λ_i), $Y = \sum_i \lambda_i z_i^2$ with $z_i \overset{\text{iid}}{\sim} N(0,1)$, so its MGF is $M_Y(t) = \prod_i(1-2\lambda_i t)^{-1/2}$ for $|t|$ small. If $Y \sim \chi^2(m)$, $\prod_i(1-2\lambda_i t) = (1-2t)^m$ near $t = 0$, hence for all t (polynomials). Comparing roots: every nonzero λ_i equals 1 and there are exactly m of them. So $\mathbf{A}$'s eigenvalues are $0/1$ with m ones, i.e. $\mathbf{A}^2 = \boldsymbol{\Gamma}\boldsymbol{\Lambda}^2\boldsymbol{\Gamma}^\top = \mathbf{A}$ is idempotent of rank m.

(c) $\boldsymbol{\Sigma} \succ 0$ has a symmetric square root, and $\mathbf{X} \overset{d}{=} \boldsymbol{\Sigma}^{1/2}\mathbf{z}$ with $\mathbf{z} \sim N_p(\mathbf{0}, \mathbf{I})$, so $Y = \mathbf{z}^\top\mathbf{B}\mathbf{z}$, $\mathbf{B} = \boldsymbol{\Sigma}^{1/2}\mathbf{A}\boldsymbol{\Sigma}^{1/2}$ symmetric. By (b) with $\sigma = 1$, $Y \sim \chi^2(m) \iff \mathbf{B}$ is idempotent of rank m. Now

$$\mathbf{B}^2 = \mathbf{B} \iff \boldsymbol{\Sigma}^{1/2}\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}\boldsymbol{\Sigma}^{1/2} = \boldsymbol{\Sigma}^{1/2}\mathbf{A}\boldsymbol{\Sigma}^{1/2} \iff \mathbf{A}\boldsymbol{\Sigma}\mathbf{A} = \mathbf{A} \iff (\mathbf{A}\boldsymbol{\Sigma})^2 = \mathbf{A}\boldsymbol{\Sigma} \iff (\boldsymbol{\Sigma}\mathbf{A})^2 = \boldsymbol{\Sigma}\mathbf{A},$$

multiplying by the invertible $\boldsymbol{\Sigma}^{\pm 1/2}$ and $\boldsymbol{\Sigma}^{\pm 1}$ on the appropriate sides (the last step transposes). Ranks agree because multiplication by invertible matrices preserves rank: $\mathrm{rank}\mathbf{B} = \mathrm{rank}\mathbf{A} = \mathrm{rank}(\mathbf{A}\boldsymbol{\Sigma}) = \mathrm{rank}(\boldsymbol{\Sigma}\mathbf{A})$. Hence $Y \sim \chi^2(m)$ iff $\mathbf{A}\boldsymbol{\Sigma}$ (equivalently $\boldsymbol{\Sigma}\mathbf{A}$) is idempotent of rank m.

Solution 16 [Ch. 4 — Wishart moments]

(a) Theorem 4.2(i) with $\mathbf{B} = \mathbf{a}$: $\mathbf{a}^\top \mathbf{M}\mathbf{a} \sim W_1(m, \mathbf{a}^\top \boldsymbol{\Sigma}\mathbf{a}) = (\mathbf{a}^\top \boldsymbol{\Sigma}\mathbf{a})\chi^2_m$. Directly, $\mathbf{a}^\top \mathbf{M}\mathbf{a} = \sum_j (\mathbf{a}^\top \mathbf{Z}_j)^2$ with $\mathbf{a}^\top \mathbf{Z}_j \overset{\text{iid}}{\sim} N(0, \mathbf{a}^\top \boldsymbol{\Sigma}\mathbf{a})$. Since $\operatorname{Var}(\chi^2_m) = 2m$, $\operatorname{Var}(\mathbf{a}^\top \mathbf{M}\mathbf{a}) = 2m(\mathbf{a}^\top \boldsymbol{\Sigma}\mathbf{a})^2$.
(b) With $\mathbf{a} = \mathbf{e}_i$, $\operatorname{Var}(M_{ii}) = 2m\Sigma_{ii}^2$, and Theorem 4.2(iv) gives $m(\Sigma_{ii}^2 + \Sigma_{ii}\Sigma_{ii}) = 2m\Sigma_{ii}^2$.

Solution 17 [Ch. 4 — the inverse-Wishart ratio theorem (proof)]

Write $\mathbf{M} = \mathbf{X}^\top\mathbf{X}$, $\mathbf{X} = [\mathbf{x}_1, \mathbf{X}_2] \in \mathbb{R}^{m\times p}$ with iid $N_p(\mathbf{0}, \mathbf{\Sigma})$ rows; $\mathbf{x}_1$ is the first column, $\mathbf{X}_2$ the other $p-1$. Partition $\mathbf{M}$ and $\mathbf{\Sigma}$ accordingly. **Step 1 (block inverse).** $\frac{1}{(\mathbf{M}^{-1})_{11}} = m_{11} - \mathbf{M}_{12}\mathbf{M}_{22}^{-1}\mathbf{M}_{21} = \mathbf{x}_1^\top\mathbf{A}\mathbf{x}_1$ with $\mathbf{A} = \mathbf{I}_m - \mathbf{X}_2(\mathbf{X}_2^\top\mathbf{X}_2)^{-1}\mathbf{X}_2^\top$, and $\frac{1}{(\mathbf{\Sigma}^{-1})_{11}} = \sigma_{11} - \boldsymbol{\sigma}_{12}\mathbf{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21} = \sigma_{11\cdot 2}$. ($\mathbf{X}_2^\top\mathbf{X}_2$ is invertible a.s. since $m \geq p-1$.) **Step 2 (conditional distribution).** Rows are independent and by Theorem 3.11, $x_{i1} \mid \mathbf{x}_{i2} \sim N(\mathbf{x}_{i2}^\top\mathbf{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21}, \sigma_{11\cdot 2})$, so given $\mathbf{X}_2$: $\mathbf{x}_1 = \mathbf{X}_2\mathbf{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21} + \mathbf{e}$, $\mathbf{e} \sim N_m(\mathbf{0}, \sigma_{11\cdot 2}\mathbf{I}_m)$. **Step 3 (projection).** $\mathbf{A}$ is the projection onto the orthogonal complement of the column space of $\mathbf{X}_2$, so $\mathbf{A}\mathbf{X}_2 = \mathbf{0}$ and $\mathbf{x}_1^\top\mathbf{A}\mathbf{x}_1 = \mathbf{e}^\top\mathbf{A}\mathbf{e}$. $\mathbf{A}$ is symmetric idempotent with $\mathrm{rank}\mathbf{A} = \mathrm{tr}\mathbf{A} = m - \mathrm{tr}\{(\mathbf{X}_2^\top\mathbf{X}_2)^{-1}\mathbf{X}_2^\top\mathbf{X}_2\} = m - (p-1)$. **Step 4.** By Theorem 3.20, $\mathbf{e}^\top\mathbf{A}\mathbf{e}/\sigma_{11\cdot 2} \mid \mathbf{X}_2 \sim \chi^2_{m-p+1}$. The conditional law does not depend on $\mathbf{X}_2$, so it is also the marginal law and the ratio is independent of $\mathbf{X}_2$. Since $(\mathbf{\Sigma}^{-1})_{11}/(\mathbf{M}^{-1})_{11} = \mathbf{x}_1^\top\mathbf{A}\mathbf{x}_1/\sigma_{11\cdot 2}$, the claim follows. **Degrees of freedom**: the residual of regressing the m-vector $\mathbf{x}_1$ on the $p-1$ columns of $\mathbf{X}_2$ lives in an $(m-p+1)$-dimensional space.

Solution 18 [Ch. 4 — inverse-Wishart ratios and $\mathrm{E}(\mathbf{S}^{-1})$]

(a) $\mathbf{M} = (n-1)\mathbf{S} \sim W_p(n-1, \boldsymbol{\Sigma})$; Theorem 4.5(ii) with $m = n-1$ gives $\frac{\mathbf{a}^\top \boldsymbol{\Sigma}^{-1}\mathbf{a}}{\mathbf{a}^\top \mathbf{M}^{-1}\mathbf{a}} = (n-1)\frac{\mathbf{a}^\top \boldsymbol{\Sigma}^{-1}\mathbf{a}}{\mathbf{a}^\top \mathbf{S}^{-1}\mathbf{a}} \sim \chi^2_{n-p}$.

(b) $\mathbf{a}^\top \mathbf{S}^{-1}\mathbf{a} = (n-1)\mathbf{a}^\top \boldsymbol{\Sigma}^{-1}\mathbf{a}/V$ with $V \sim \chi^2_{n-p}$, so $\mathrm{E}(\mathbf{a}^\top \mathbf{S}^{-1}\mathbf{a}) = \frac{n-1}{n-p-2}\mathbf{a}^\top \boldsymbol{\Sigma}^{-1}\mathbf{a}$ for all $\mathbf{a}$. For symmetric $\mathbf{A}, \mathbf{B}$, $\mathbf{a}^\top \mathbf{A}\mathbf{a} = \mathbf{a}^\top \mathbf{B}\mathbf{a}$ for all $\mathbf{a}$ implies $\mathbf{A} = \mathbf{B}$ (take $\mathbf{a} = \mathbf{e}_i$ and $\mathbf{e}_i + \mathbf{e}_j$). Applying this to $\mathrm{E}\mathbf{S}^{-1}$ and $\frac{n-1}{n-p-2}\boldsymbol{\Sigma}^{-1}$ gives the claim; $\mathbf{S}^{-1}$ is biased for $\boldsymbol{\Sigma}^{-1}$ and $\frac{n-p-2}{n-1}\mathbf{S}^{-1}$ is unbiased.

Solution 19 [Ch. 4 — constrained MLEs]

(a) $\ell(\boldsymbol{\Sigma}) = \frac{n}{2}\{\log|\boldsymbol{\Sigma}| + \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{S}_{\mu_0})\} + c$ with $\mathbf{S}_{\mu_0} = \frac{1}{n}\sum(\mathbf{x}_i - \boldsymbol{\mu}_0)(\mathbf{x}_i - \boldsymbol{\mu}_0)^\top$; by Problem 7, $\hat{\boldsymbol{\Sigma}} = \mathbf{S}_{\mu_0}$.

(b) With $\boldsymbol{\Sigma} = \sigma^2\mathbf{I}$, $\ell = \frac{np}{2}\log\sigma^2 + \frac{1}{2\sigma^2}\sum_i \|\mathbf{x}_i - \bar{\mathbf{x}}\|^2 + c$; setting the derivative to zero gives $\hat{\sigma}^2 = \frac{1}{np}\sum_i \|\mathbf{x}_i - \bar{\mathbf{x}}\|^2 = \operatorname{tr}(\mathbf{S}_0)/p$.

(c) With $\boldsymbol{\Sigma}$ diagonal the log-likelihood separates: $\ell = \frac{n}{2}\sum_j\{\log\sigma_{jj} + (\mathbf{S}_0)_{jj}/\sigma_{jj}\} + c$, minimized at $\hat{\sigma}_{jj} = (\mathbf{S}_0)_{jj}$.

Solution 20 [Ch. 4 — large-sample theory]

(a) CLT: $\mathbf{Y}_n = \sqrt{n}(\bar{\mathbf{X}} - \boldsymbol{\mu}) \xrightarrow{d} N_p(\mathbf{0}, \boldsymbol{\Sigma})$. $\mathbf{S} \xrightarrow{p} \boldsymbol{\Sigma}$ and inversion is continuous at $\boldsymbol{\Sigma}$, so $\mathbf{S}^{-1} \xrightarrow{p} \boldsymbol{\Sigma}^{-1}$. By Slutsky $(\mathbf{Y}_n, \mathbf{S}^{-1}) \xrightarrow{d} (\mathbf{Y}, \boldsymbol{\Sigma}^{-1})$ and $g(\mathbf{y}, \mathbf{A}) = \mathbf{y}^\top \mathbf{A}\mathbf{y}$ is continuous, so by CMT $\mathbf{Y}_n^\top \mathbf{S}^{-1} \mathbf{Y}_n \xrightarrow{d} \mathbf{Y}^\top \boldsymbol{\Sigma}^{-1} \mathbf{Y} \sim \chi^2_p$.

(b) $h(\boldsymbol{\mu}) = \mu_1/\mu_2$, $\nabla h = (1/\mu_2, -\mu_1/\mu_2^2)^\top$:

$$\sqrt{n}\Big(\frac{\bar{X}_1}{\bar{X}_2} - \frac{\mu_1}{\mu_2}\Big) \xrightarrow{d} N\Big(0,\ \frac{\sigma_{11}}{\mu_2^2} - \frac{2\mu_1\sigma_{12}}{\mu_2^3} + \frac{\mu_1^2\sigma_{22}}{\mu_2^4}\Big).$$

Solution 21 [Ch. 4 — asymptotics of sample covariances]

(a) Let $\mathbf{Z}_i = \mathbf{X}_i - \boldsymbol{\mu} \overset{\text{iid}}{\sim} N_p(\mathbf{0}, \boldsymbol{\Sigma})$. Then

$$s_{jk} = \frac{n}{n-1}\Big\{\frac{1}{n}\sum_i Z_{ij}Z_{ik} - \bar{Z}_j\bar{Z}_k\Big\}.$$

$\sqrt{n}\bar{Z}_j\bar{Z}_k = \sqrt{n}\bar{Z}_j \cdot \bar{Z}_k = O_p(1)o_p(1) \to 0$ and $\frac{n}{n-1} \to 1$. The terms $W_i = Z_{ij}Z_{ik}$ are iid with $\mathrm{E}W_i = \sigma_{jk}$ and, as in the proof of Theorem 4.2(iv), $\mathrm{E}W_i^2 = 2\sigma_{jk}^2 + \sigma_{jj}\sigma_{kk}$, so $\mathrm{Var}W_i = \sigma_{jk}^2 + \sigma_{jj}\sigma_{kk}$. The CLT and Slutsky give $\sqrt{n}(s_{jk} - \sigma_{jk}) \xrightarrow{d} N(0, \sigma_{jk}^2 + \sigma_{jj}\sigma_{kk})$.

(b) For $j = k$ the variance is $2\sigma_{jj}^2$; replacing σ_{jj} by s_{jj} (Slutsky), $s_{jj} \pm 1.96\, s_{jj}\sqrt{2/n}$. With $h = \log$, $h'(\sigma) = 1/\sigma$: $\sqrt{n}(\log s_{jj} - \log \sigma_{jj}) \xrightarrow{d} N(0, 2)$, so $\log s_{jj} \pm 1.96\sqrt{2/n}$; exponentiating gives an interval for σ_{jj} that stays positive.

Solution 22 [Ch. 5 — computing a T^2 test]

(a) $\bar{\mathbf{x}} = (6, 10)^\top$; deviations $(-4, 2), (2, -1), (0, -1), (2, 0)$:

$$\mathbf{S} = \begin{pmatrix} 8 & -10/3 \\ -10/3 & 2 \end{pmatrix}, \quad |\mathbf{S}| = \tfrac{44}{9}, \quad \mathbf{S}^{-1} = \frac{9}{44}\begin{pmatrix} 2 & 10/3 \\ 10/3 & 8 \end{pmatrix} = \begin{pmatrix} .409 & .682 \\ .682 & 1.636 \end{pmatrix}.$$

$\bar{\mathbf{x}} - \boldsymbol{\mu}_0 = (-1, -1)^\top$, so $T^2 = 4 \cdot \frac{18+60+72}{44} = 13.64$. Critical value $\frac{(n-1)p}{n-p} F_{2,2}(.05) = 3 \times 19 = 57$; since $13.64 < 57$, do not reject H_0. $F_{\text{obs}} = \frac{2}{6}\,13.64 = 4.545$ and $P(F_{2,2} \geq x) = 1/(1+x)$, so $p = 0.180$.

(b) $\hat{\boldsymbol{\Sigma}} = \frac{3}{4}\mathbf{S}$, $|\hat{\boldsymbol{\Sigma}}| = \frac{9}{16} \cdot \frac{44}{9} = 2.75$. $\hat{\boldsymbol{\Sigma}}_0 = \hat{\boldsymbol{\Sigma}} + \boldsymbol{\delta}\boldsymbol{\delta}^\top$ with $\boldsymbol{\delta}^\top \hat{\boldsymbol{\Sigma}}^{-1} \boldsymbol{\delta} = \frac{4}{3} \cdot \frac{150}{44} = 4.545$, so $|\hat{\boldsymbol{\Sigma}}_0| = 2.75 \times 5.545 = 15.25$ and $\Lambda^{2/n} = 2.75/15.25 = 0.1803 = (1 + 13.64/3)^{-1}$.

(c) T^2: $c = \sqrt{57} = 7.55$; μ_1: $6 \pm 7.55\sqrt{8/4} = (-4.68, 16.68)$; μ_2: $10 \pm 7.55\sqrt{2/4} = (4.66, 15.34)$. Bonferroni: $t_3(.05/4) = 4.177$; μ_1: $6 \pm 4.177\sqrt{2} = (0.09, 11.91)$; μ_2: $10 \pm 4.177\sqrt{.5} = (7.05, 12.95)$. Bonferroni is much shorter here ($n = 4$, heavy-tailed $F_{2,2}$). Both contain $(7, 11)$, consistent with the test.

Solution 23 [Ch. 5 — likelihood ratio tests for the mean]

(a) $\sum_j (\mathbf{x}_j - \boldsymbol{\mu}_0)(\mathbf{x}_j - \boldsymbol{\mu}_0)^\top = \sum_j (\mathbf{x}_j - \bar{\mathbf{x}})(\mathbf{x}_j - \bar{\mathbf{x}})^\top + n\boldsymbol{\delta}\boldsymbol{\delta}^\top$ ($\boldsymbol{\delta} = \bar{\mathbf{x}} - \boldsymbol{\mu}_0$, cross terms vanish), i.e. $\hat{\boldsymbol{\Sigma}}_0 = \hat{\boldsymbol{\Sigma}} + \boldsymbol{\delta}\boldsymbol{\delta}^\top$. Problem 8(a) gives $|\hat{\boldsymbol{\Sigma}}_0| = |\hat{\boldsymbol{\Sigma}}|(1 + \boldsymbol{\delta}^\top \hat{\boldsymbol{\Sigma}}^{-1}\boldsymbol{\delta})$, and $\hat{\boldsymbol{\Sigma}} = \frac{n-1}{n}\mathbf{S}$ gives $\boldsymbol{\delta}^\top \hat{\boldsymbol{\Sigma}}^{-1}\boldsymbol{\delta} = \frac{T^2}{n-1}$.

(b) $-2\log\Lambda = \sum_j (\mathbf{x}_j - \boldsymbol{\mu}_0)^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}_j - \boldsymbol{\mu}_0) - \sum_j (\mathbf{x}_j - \bar{\mathbf{x}})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}_j - \bar{\mathbf{x}}) = n(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)^\top \boldsymbol{\Sigma}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)$; under H_0, $\sqrt{n}\boldsymbol{\Sigma}^{-1/2}(\bar{\mathbf{X}} - \boldsymbol{\mu}_0) \sim N_p(\mathbf{0}, \mathbf{I})$, so it is exactly χ^2_p.

(c) $-2\log\Lambda = n\log(1 + T^2/(n-1))$. Under H_0, $T^2 \xrightarrow{d} \chi^2_p$ (Problem 20(a)), so $T^2 = O_p(1)$ and $\log(1+x) = x + O(x^2)$ gives $-2\log\Lambda = \frac{n}{n-1}T^2 + O_p(1/n) \xrightarrow{d} \chi^2_p$.

Solution 24 [Ch. 5 — likelihood ratio test for $H_0 : \boldsymbol{\Sigma} = \boldsymbol{\Sigma}_0$]

(a) $-2\log L(\boldsymbol{\mu},\boldsymbol{\Sigma}) = n\log|\boldsymbol{\Sigma}| + n(\bar{\mathbf{x}}-\boldsymbol{\mu})^\top\boldsymbol{\Sigma}^{-1}(\bar{\mathbf{x}}-\boldsymbol{\mu}) + n\mathrm{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{S}_0) + c$. Under H_1: $(\hat{\boldsymbol{\mu}},\hat{\boldsymbol{\Sigma}}) = (\bar{\mathbf{x}},\mathbf{S}_0)$, $-2\log L_1 = n\log|\mathbf{S}_0| + np + c$. Under H_0: $\hat{\boldsymbol{\mu}} = \bar{\mathbf{x}}$, $-2\log L_0 = n\log|\boldsymbol{\Sigma}_0| + n\mathrm{tr}(\boldsymbol{\Sigma}_0^{-1}\mathbf{S}_0) + c$. Subtracting,

$$W = n\log|\boldsymbol{\Sigma}_0| - n\log|\mathbf{S}_0| + n\mathrm{tr}(\boldsymbol{\Sigma}_0^{-1}\mathbf{S}_0) - np = -n\log|\boldsymbol{\Sigma}_0^{-1}\mathbf{S}_0| - np + n\mathrm{tr}(\boldsymbol{\Sigma}_0^{-1}\mathbf{S}_0).$$

(b) $\boldsymbol{\Sigma}_0^{-1}\mathbf{S}_0$ is similar to $\boldsymbol{\Sigma}_0^{-1/2}\mathbf{S}_0\boldsymbol{\Sigma}_0^{-1/2} \succ 0$, so its eigenvalues l_i are positive, $|\boldsymbol{\Sigma}_0^{-1}\mathbf{S}_0| = \prod l_i$, $\mathrm{tr} = \sum l_i$, and $W = n\sum_i(l_i - \log l_i - 1) \geq 0$ (Theorem 2.14), with equality iff all $l_i = 1$, i.e. $\boldsymbol{\Sigma}_0^{-1/2}\mathbf{S}_0\boldsymbol{\Sigma}_0^{-1/2} = \mathbf{I}$, i.e. $\mathbf{S}_0 = \boldsymbol{\Sigma}_0$.

(c) $p = 1$: $W = n(l - \log l - 1)$ with $l = s_0^2/\sigma_0^2$. Under H_0, $ns_0^2/\sigma_0^2 \sim \chi^2_{n-1}$, so $l \xrightarrow{p} 1$ and $\sqrt{n}(l-1) \xrightarrow{d} N(0,2)$ (CLT for $\sum Z_i^2$, $\mathrm{Var}Z^2 = 2$). Taylor: $l - \log l - 1 = \frac{1}{2}(l-1)^2 + O(|l-1|^3)$, so

$$W = \frac{1}{2}\{\sqrt{n}(l-1)\}^2 + O_p(n^{-1/2}) \xrightarrow{d} \frac{1}{2}(\sqrt{2}Z)^2 = Z^2 \sim \chi^2_1.$$

(d) H_0 fixes all free parameters of the symmetric $\boldsymbol{\Sigma}$, i.e. $p(p+1)/2$ of them. Directly (Ch. 5): after reducing to $\boldsymbol{\Sigma}_0 = \mathbf{I}$, $W \approx \frac{n}{2}\|\mathbf{S}_0 - \mathbf{I}\|_F^2 = \sum_j \frac{\{\sqrt{n}(s_{jj}-1)\}^2}{2} + \sum_{j<k}\{\sqrt{n}s_{jk}\}^2$, whose $p + p(p-1)/2$ terms converge to independent χ^2_1 variables.

Solution 25 [Ch. 5 — simultaneous intervals are shadows of the ellipsoid]

(a) Write $\boldsymbol{\mu} = \bar{\mathbf{x}} - \mathbf{v}$, so $E = \{n\mathbf{v}^\top \mathbf{S}^{-1}\mathbf{v} \le c^2\}$. With $\mathbf{w} = \sqrt{n}\mathbf{S}^{-1/2}\mathbf{v}$ ($\|\mathbf{w}\| \le c$), $\mathbf{a}^\top\mathbf{v} = \mathbf{a}^\top\mathbf{S}^{1/2}\mathbf{w}/\sqrt{n} \le \|\mathbf{S}^{1/2}\mathbf{a}\|c/\sqrt{n} = c\sqrt{\mathbf{a}^\top\mathbf{S}\mathbf{a}/n}$ by Cauchy–Schwarz, with equality at $\mathbf{w} \propto \mathbf{S}^{1/2}\mathbf{a}$, $\|\mathbf{w}\| = c$. So $\mathbf{a}^\top\boldsymbol{\mu}$ ranges exactly over $\mathbf{a}^\top\bar{\mathbf{x}} \pm c\sqrt{\mathbf{a}^\top\mathbf{S}\mathbf{a}/n}$.

(b) If $\boldsymbol{\mu} \in E$, (a) shows every interval contains $\mathbf{a}^\top\boldsymbol{\mu}$. If $\boldsymbol{\mu} \notin E$, take $\mathbf{a} = \mathbf{S}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu})$: $t(\mathbf{a})^2 = T^2 > c^2$ and that interval misses. So $\{\text{all intervals hold}\} = \{\boldsymbol{\mu} \in E\} = \{T^2 \le c^2\}$, of probability exactly $1 - \alpha$. Componentwise intervals are the shadows of E on the axes.

Solution 26 [Ch. 5 — Bonferroni versus T^2]

(i) $t_{19}(.025) = 2.093$; (ii) $t_{19}(.05/6) = 2.625$; (iii) $\sqrt{\frac{3\cdot 19}{17} F_{3,17}(.05)} = \sqrt{3.353 \times 3.197} = 3.274$. Ratio $2.625/3.274 = 0.802$. (i) is for a single pre-specified comparison, (ii) for a few pre-specified ones, (iii) for all linear combinations including those chosen after looking at the data. Using (i) three times can drop the joint confidence to $1 - 3(0.05) = 0.85$.

Solution 27 [Ch. 6 — repeated measures design]

$\bar{\mathbf{y}} = (10, 12.167, 14.167)^\top$; differences $\mathbf{D}_j = \mathbf{C}\mathbf{y}_j$: $(2,5), (3,3), (2,5), (0,3), (3,5), (3,4)$, so

$$\bar{\mathbf{D}} = \mathbf{C}\bar{\mathbf{y}} = \left(\frac{13}{6}, \frac{25}{6}\right)^\top, \qquad \mathbf{S}_d = \mathbf{C}\mathbf{S}\mathbf{C}^\top = \frac{1}{30}\begin{pmatrix} 41 & 11 \\ 11 & 29 \end{pmatrix}, \qquad |\mathbf{S}_d| = \frac{1068}{900}.$$

$T^2 = 6\,\bar{\mathbf{D}}^\top \mathbf{S}_d^{-1} \bar{\mathbf{D}} = 6 \cdot \frac{30}{1068} \cdot \frac{29 \cdot 169 - 2 \cdot 11 \cdot 325 + 41 \cdot 625}{36} = 109.4$. Critical value with $q = 2$: $\frac{5 \cdot 2}{4} F_{2,4}(.05) = 2.5 \times 6.944 = 17.36$; reject. Since $\mathbf{C}\mathbf{S}\mathbf{C}^\top$ is exactly the sample covariance of the $\mathbf{D}_j$, the contrast test is the one-sample T^2 for $H_0 : \mathrm{E}\mathbf{D} = \mathbf{0}$.

Solution 28 [Ch. 6 — two-sample T^2]

(a) $\mathbf{d} = (-1.9, 0.2)^\top$, $\mathbf{S}_{\text{pooled}}^{-1}\mathbf{d} = (-1.078, 0.256)^\top$, $\mathbf{d}^\top\mathbf{S}_{\text{pooled}}^{-1}\mathbf{d} = 2.099$, $T^2 = 2.099/0.04 = 52.47$. $c^2 = \frac{98\cdot 2}{97}F_{2,97}(.05) = 2.021 \times 3.09 = 6.24$: reject.

(b) $c = 2.499$. Component 1: $-1.9 \pm 2.499\sqrt{0.04 \cdot 2} = (-2.61, -1.19)$; component 2: $0.2 \pm 2.499\sqrt{0.04 \cdot 5} = (-0.92, 1.32)$. Only the first excludes 0: the difference comes from variable 1 (consistent with the large first entry of $\mathbf{S}_{\text{pooled}}^{-1}\mathbf{d}$).

(c) $\Lambda^* = (1 + T^2/(n_1 + n_2 - 2))^{-1} = (1 + 52.47/98)^{-1} = 0.651$.

Solution 29 [Ch. 6 — one-way MANOVA by hand]

Group means $(8,4)$, $(1,2)$, $(2,8)$; grand mean $(4,5)$; $n=8$.

$$\mathbf{B} = 3\begin{pmatrix}16 & -4\\ -4 & 1\end{pmatrix} + 2\begin{pmatrix}9 & 9\\ 9 & 9\end{pmatrix} + 3\begin{pmatrix}4 & -6\\ -6 & 9\end{pmatrix} = \begin{pmatrix}78 & -12\\ -12 & 48\end{pmatrix},$$

$$\mathbf{W} = \begin{pmatrix}6 & 6\\ 6 & 14\end{pmatrix} + \begin{pmatrix}2 & -4\\ -4 & 8\end{pmatrix} + \begin{pmatrix}2 & -1\\ -1 & 2\end{pmatrix} = \begin{pmatrix}10 & 1\\ 1 & 24\end{pmatrix}.$$

$|\mathbf{W}| = 239$, $|\mathbf{B}+\mathbf{W}| = \begin{vmatrix}88 & -11\\ -11 & 72\end{vmatrix} = 6215$, $\Lambda^* = 0.0385$. Exact ($p=2$): $\frac{n-g-1}{g-1}\cdot\frac{1-\sqrt{\Lambda^*}}{\sqrt{\Lambda^*}} = 2\cdot\frac{1-0.1961}{0.1961} = 8.20 > F_{4,8}(.01) = 7.01$: reject. Bartlett: $-(8-1-\frac{5}{2})\ln 0.0385 = 4.5\times 3.258 = 14.66 > \chi^2_4(.01) = 13.28$: reject ($n$ is tiny, so this is only indicative). Eigenvalues of $\mathbf{W}^{-1}\mathbf{B}$: $\eta_1 = 8.076$, $\eta_2 = 1.865$; $\prod(1+\eta_i)^{-1} = 1/(9.076\times 2.865) = 0.0385$.

Solution 30 [Ch. 6 — expectations of $\mathbf{B}$ and $\mathbf{W}$]

$\mathrm{E}\{(n_\ell-1)\mathbf{S}_\ell\} = (n_\ell-1)\boldsymbol{\Sigma}$, so $\mathrm{E}\mathbf{W} = (n-g)\boldsymbol{\Sigma}$. With $\bar{\mathbf{e}}_\ell$ the error mean of group ℓ and $\bar{\mathbf{e}} = \sum n_\ell \bar{\mathbf{e}}_\ell / n$, $\sum n_\ell \boldsymbol{\tau}_\ell = \mathbf{0}$ gives $\bar{\mathbf{x}}_\ell - \bar{\mathbf{x}} = \boldsymbol{\tau}_\ell + (\bar{\mathbf{e}}_\ell - \bar{\mathbf{e}})$ and

$$\mathrm{E}\mathbf{B} = \sum_\ell n_\ell \boldsymbol{\tau}_\ell \boldsymbol{\tau}_\ell^\top + \mathrm{E}\Big[\sum_\ell n_\ell \bar{\mathbf{e}}_\ell \bar{\mathbf{e}}_\ell^\top - n\bar{\mathbf{e}}\bar{\mathbf{e}}^\top\Big] = \sum_\ell n_\ell \boldsymbol{\tau}_\ell \boldsymbol{\tau}_\ell^\top + \sum_\ell n_\ell \frac{\boldsymbol{\Sigma}}{n_\ell} - n\frac{\boldsymbol{\Sigma}}{n}$$

(cross terms have mean zero), i.e. $(g-1)\boldsymbol{\Sigma} + \sum n_\ell \boldsymbol{\tau}_\ell \boldsymbol{\tau}_\ell^\top$. Under H_0 both $\mathbf{B}/(g-1)$ and $\mathbf{W}/(n-g)$ estimate $\boldsymbol{\Sigma}$; under H_1, $\mathbf{B}$ gains a positive semidefinite term. Small $|\mathbf{W}|/|\mathbf{B}+\mathbf{W}|$ (large eigenvalues of $\mathbf{W}^{-1}\mathbf{B}$) is evidence against H_0.

Solution 31 [Ch. 6 — Wilks' lambda is the LRT]

(a) The log-likelihood is $-\frac{n}{2}\log|\boldsymbol{\Sigma}| - \frac{1}{2}\sum_\ell\sum_j(\mathbf{x}_{\ell j}-\boldsymbol{\mu}_\ell)^\top\boldsymbol{\Sigma}^{-1}(\mathbf{x}_{\ell j}-\boldsymbol{\mu}_\ell) + c$. Under H_1 (free $\boldsymbol{\mu}_\ell$) the quadratic term is minimized by $\hat{\boldsymbol{\mu}}_\ell = \bar{\mathbf{x}}_\ell$, leaving $-\frac{n}{2}\{\log|\boldsymbol{\Sigma}| + \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{W}/n)\}$, maximized at $\hat{\boldsymbol{\Sigma}} = \mathbf{W}/n$ (Problem 7). Under H_0 (common $\boldsymbol{\mu}$), $\hat{\boldsymbol{\mu}} = \bar{\mathbf{x}}$ and $\hat{\boldsymbol{\Sigma}}_0 = \mathbf{T}/n = (\mathbf{B}+\mathbf{W})/n$. The maximized likelihoods are $(2\pi)^{-np/2}|\hat{\boldsymbol{\Sigma}}|^{-n/2}e^{-np/2}$, so $\Lambda = (|\mathbf{W}|/|\mathbf{B}+\mathbf{W}|)^{n/2}$ and $\Lambda^{2/n} = \Lambda^*$.

(b) H_1 has g mean vectors (gp parameters), H_0 one (p), with $\boldsymbol{\Sigma}$ free in both: the difference is $p(g-1)$, the degrees of freedom of $-2\log\Lambda = -n\ln\Lambda^*$; Bartlett replaces n by $n - 1 - \frac{p+g}{2}$.

(c) For $g=2$, $\mathbf{B} = \frac{n_1 n_2}{n}\mathbf{d}\mathbf{d}^\top$ ($\mathbf{d} = \bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2$, since $\bar{\mathbf{x}}_1 - \bar{\mathbf{x}} = \frac{n_2}{n}\mathbf{d}$, $\bar{\mathbf{x}}_2 - \bar{\mathbf{x}} = -\frac{n_1}{n}\mathbf{d}$). By Problem 8(c), $\Lambda^* = (1 + \frac{n_1 n_2}{n}\mathbf{d}^\top\mathbf{W}^{-1}\mathbf{d})^{-1} = (1 + \frac{T^2}{n-2})^{-1}$ because $\mathbf{W} = (n-2)\mathbf{S}_{\text{pooled}}$. Hence $\frac{1-\Lambda^*}{\Lambda^*} = \frac{T^2}{n-2}$ and Theorem 6.5 gives $\frac{n-p-1}{p}\cdot\frac{1-\Lambda^*}{\Lambda^*} \sim F_{p,n-p-1}$.

Solution 32 [Ch. 6 — two populations with unequal covariances]

$N = n_1 + n_2$, $\boldsymbol{\delta} = \bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2 - (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$, $\mathbf{V}_N = \frac{\boldsymbol{\Sigma}_1}{n_1} + \frac{\boldsymbol{\Sigma}_2}{n_2}$, $\tilde{\mathbf{V}}_N = \frac{\mathbf{S}_1}{n_1} + \frac{\mathbf{S}_2}{n_2}$. Then $N\mathbf{V}_N \to \mathbf{V} := \boldsymbol{\Sigma}_1/\kappa + \boldsymbol{\Sigma}_2/(1-\kappa) \succ 0$ and, since $\mathbf{S}_i \xrightarrow{p} \boldsymbol{\Sigma}_i$, $N\tilde{\mathbf{V}}_N \xrightarrow{p} \mathbf{V}$. Also $\sqrt{N}\boldsymbol{\delta} \sim N_p(\mathbf{0}, N\mathbf{V}_N) \xrightarrow{d} N_p(\mathbf{0}, \mathbf{V})$ (without normality, by the CLT in each independent sample). By Slutsky and CMT,

$$\boldsymbol{\delta}^\top \tilde{\mathbf{V}}_N^{-1} \boldsymbol{\delta} = (\sqrt{N}\boldsymbol{\delta})^\top (N\tilde{\mathbf{V}}_N)^{-1} (\sqrt{N}\boldsymbol{\delta}) \xrightarrow{d} \mathbf{Z}^\top \mathbf{V}^{-1} \mathbf{Z} \sim \chi^2_p, \qquad \mathbf{Z} \sim N_p(\mathbf{0}, \mathbf{V}).$$

Part II. Dimension Reduction, Classification, Clustering

Solution 33 [Ch. 7 — computing principal components]

(a) X_3 is uncorrelated with the others, so $\lambda = 2$ with $\mathbf{u} = \mathbf{e}_3$. The block $\begin{pmatrix} 1 & -2 \\ -2 & 5 \end{pmatrix}$ has eigenvalues $3 \pm 2\sqrt{2}$:

$\lambda_1 = 5.83,\ \mathbf{u}_1 = (-0.383, 0.924, 0)^\top; \quad \lambda_2 = 2.00,\ \mathbf{u}_2 = (0, 0, 1)^\top; \quad \lambda_3 = 0.17,\ \mathbf{u}_3 = (0.924, 0.383, 0)^\top.$

Proportions: $5.83/8 = 72.9\%$, cumulative 97.9%.

(b) $\text{Corr}(Z_1, X_1) = u_{11}\sqrt{\lambda_1}/\sqrt{\sigma_{11}} = -0.383 \times 2.414 = -0.924$, $\text{Corr}(Z_1, X_2) = 0.924 \times 2.414/\sqrt{5} = 0.998$: Z_1 is essentially X_2.

(c) $\mathbf{R} = \begin{pmatrix} 1 & -0.894 & 0 \\ -0.894 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ has eigenvalues $1.894, 1, 0.106$ and $\mathbf{u}_1^R = (0.707, -0.707, 0)^\top$: unlike covariance PCA (dominated by the large-variance X_2), X_1 and X_2 get equal weight; proportion $1.894/3 = 63.1\%$. The result depends strongly on scale.

Solution 34 [2024 Midterm Q3 — PCA of a singular covariance]

$\det(\boldsymbol{\Sigma} - \lambda \mathbf{I}) = (1-\lambda)^2 - 1 = 0$ gives $\lambda_1 = 2$, $\lambda_2 = 0$, with $\mathbf{u}_1 = (1, -1)^\top / \sqrt{2}$. The first PC $Z_1 = (Y_1 - Y_2)/\sqrt{2}$ explains $\lambda_1/(\lambda_1 + \lambda_2) = 2/2 = 100\%$ of the total variance. ($Y_1 + Y_2$ has variance $1 + 1 - 2 = 0$: $Y_2 = -Y_1 + \text{const}$ a.s.)

Solution 35 [Ch. 7 — equicorrelation]

$\boldsymbol{\Sigma}\mathbf{1} = \sigma^2\{(1-\rho)+p\rho\}\mathbf{1}$, so $\lambda_1 = \sigma^2\{1+(p-1)\rho\}$ with $\mathbf{u}_1 = \mathbf{1}/\sqrt{p}$. For $\mathbf{v} \perp \mathbf{1}$, $\boldsymbol{\Sigma}\mathbf{v} = \sigma^2(1-\rho)\mathbf{v}$: the other $p-1$ eigenvalues all equal $\sigma^2(1-\rho)$ (any orthonormal basis of $\mathbf{1}^\perp$; not unique). $Z_1 = \frac{1}{\sqrt{p}}\sum X_j$ measures overall size, with proportion $\frac{1+(p-1)\rho}{p}$: $\to 1$ as $\rho \to 1$ (one component explains everything) and $= 1/p$ at $\rho = 0$ (no gain from reduction).

Solution 36 [Ch. 7 — reconstruction error and SVD]

(a) With $\mathbf{y}_i = \mathbf{x}_i - \bar{\mathbf{x}}$ and $\mathbf{P} = \mathbf{V}\mathbf{V}^\top$ symmetric idempotent, $\|\mathbf{y}_i - \mathbf{P}\mathbf{y}_i\|^2 = \mathbf{y}_i^\top\mathbf{y}_i - \mathbf{y}_i^\top\mathbf{V}\mathbf{V}^\top\mathbf{y}_i$. Summing, $\sum \mathbf{y}_i^\top\mathbf{y}_i = (n-1)\mathrm{tr}\mathbf{S}$ and $\sum \mathbf{y}_i^\top\mathbf{V}\mathbf{V}^\top\mathbf{y}_i = \mathrm{tr}(\mathbf{V}^\top\sum \mathbf{y}_i\mathbf{y}_i^\top\mathbf{V}) = (n-1)\mathrm{tr}(\mathbf{V}^\top\mathbf{S}\mathbf{V})$. By von Neumann, $\mathrm{tr}(\mathbf{V}^\top\mathbf{S}\mathbf{V}) = \mathrm{tr}(\mathbf{S}\mathbf{P}) \le \sum \lambda_i(\mathbf{S})\lambda_i(\mathbf{P}) = \hat{\lambda}_1 + \cdots + \hat{\lambda}_q$ ($\mathbf{P}$ has q unit eigenvalues), with equality at $\mathbf{V} = [\hat{\mathbf{u}}_1, \ldots, \hat{\mathbf{u}}_q]$. The minimum error is $(n-1)\sum_{j>q}\hat{\lambda}_j$.

(b) $\mathbf{S} = \frac{1}{n-1}\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}} = \frac{1}{n-1}\mathbf{U}\mathbf{D}\mathbf{V}^\top\mathbf{V}\mathbf{D}\mathbf{U}^\top = \mathbf{U}\frac{\mathbf{D}^2}{n-1}\mathbf{U}^\top$, so $\mathbf{U}$ holds the PC directions; the score matrix $\mathbf{T} = \tilde{\mathbf{X}}\mathbf{U}$ has transpose $\mathbf{U}^\top\tilde{\mathbf{X}}^\top = \mathbf{U}^\top\mathbf{U}\mathbf{D}\mathbf{V}^\top = \mathbf{D}\mathbf{V}^\top$.

Solution 37 [Ch. 8 — computing canonical correlations]

The singular values of $\mathbf{M} = \mathbf{R}_{11}^{-1/2}\mathbf{R}_{12}\mathbf{R}_{22}^{-1/2}$ are $\rho_1^* = 0.739$, $\rho_2^* = 0.030$. Normalized first coefficients: $\mathbf{g}_1 = \mathbf{R}_{11}^{-1/2}\mathbf{a}_1 = (0.856, 0.278)^\top$, $\mathbf{h}_1 = \mathbf{R}_{22}^{-1/2}\mathbf{b}_1 = (0.545, 0.737)^\top$, with $\mathbf{g}_1^\top \mathbf{R}_{11}\mathbf{g}_1 = \mathbf{h}_1^\top \mathbf{R}_{22}\mathbf{h}_1 = 1$ and $\mathbf{g}_1^\top \mathbf{R}_{12}\mathbf{h}_1 = 0.739$. Check: $(1 - 0.739^2)(1 - 0.030^2) = 0.4539 = 0.3660/(0.84 \times 0.96) = |\mathbf{R}|/(|\mathbf{R}_{11}||\mathbf{R}_{22}|)$. The association is essentially one-dimensional.

Solution 38 [Ch. 8 — properties of canonical correlations]

(a) For $q = 1$, $\mathbf{M} = \boldsymbol{\Sigma}_{11}^{-1/2}\boldsymbol{\sigma}_{12}\sigma_{22}^{-1/2}$ is a vector; the only nonzero eigenvalue of $\mathbf{M}\mathbf{M}^\top$ is $\mathbf{M}^\top\mathbf{M} = \boldsymbol{\sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1}\boldsymbol{\sigma}_{12}/\sigma_{22} = \rho^2_{\text{mult}}$.

(b) $\operatorname{Corr}(\mathbf{g}^\top(\mathbf{A}\mathbf{X}+\mathbf{b}), \mathbf{h}^\top(\mathbf{C}\mathbf{Y}+\mathbf{d})) = \operatorname{Corr}((\mathbf{A}^\top\mathbf{g})^\top\mathbf{X}, (\mathbf{C}^\top\mathbf{h})^\top\mathbf{Y})$ and $\mathbf{g} \mapsto \mathbf{A}^\top\mathbf{g}$, $\mathbf{h} \mapsto \mathbf{C}^\top\mathbf{h}$ are bijections, also mapping the uncorrelatedness constraints onto each other; hence all ρ_k agree. Constants do not affect correlations.

(c) $\operatorname{Cov}(\xi_k, \omega_l) = \mathbf{g}_k^\top\boldsymbol{\Sigma}_{12}\mathbf{h}_l = \mathbf{a}_k^\top\mathbf{M}\mathbf{b}_l = \mathbf{a}_k^\top(d_l\mathbf{a}_l) = d_l\delta_{kl}$, using $\mathbf{M}\mathbf{b}_l = d_l\mathbf{a}_l$.

Solution 39 [Ch. 9 — the factor model]

(a)

$$\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi} = \begin{pmatrix} 19 & 30 & 2 & 12 \\ 30 & 57 & 5 & 23 \\ 2 & 5 & 38 & 47 \\ 12 & 23 & 47 & 68 \end{pmatrix}, \qquad h_i^2 = 17, 53, 37, 65,$$

e.g. $\sigma_{11} = 19 = (4^2 + 1^2) + 2$. $(\boldsymbol{\Lambda}\mathbf{T})(\boldsymbol{\Lambda}\mathbf{T})^\top = \boldsymbol{\Lambda}\mathbf{T}\mathbf{T}^\top\boldsymbol{\Lambda}^\top = \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top$.

(b) $\lambda_1\lambda_2 = .9$, $\lambda_1\lambda_3 = .7$, $\lambda_2\lambda_3 = .4$ give $\lambda_1^2 = .9 \times .7/.4 = 1.575$, $\lambda_1 = 1.255$, $\lambda_2 = .717$, $\lambda_3 = .558$; then $\psi_1 = 1 - 1.575 = -0.575 < 0$ (a negative variance) and $\text{Corr}(X_1, F_1) = \lambda_1 > 1$: impossible. The model fits exactly but the solution is improper — a **Heywood case**. ($\psi_2 = .486$, $\psi_3 = .689$.)

(c) $\nu = \frac{p(p+1)}{2} - pk - p + \frac{k(k-1)}{2} = \frac{p^2 - 2pk + k^2 - p - k}{2} = \frac{(p-k)^2 - p - k}{2}$. For $p = 3$, $k = 1$, $\nu = 0$: an exact fit with a unique solution, as in (b).

Solution 40 [Ch. 9 — factor scores]

$Q(\mathbf{F}) = (\mathbf{x}-\boldsymbol{\mu}-\boldsymbol{\Lambda}\mathbf{F})^\top\boldsymbol{\Psi}^{-1}(\mathbf{x}-\boldsymbol{\mu}-\boldsymbol{\Lambda}\mathbf{F})$ has gradient $-2\boldsymbol{\Lambda}^\top\boldsymbol{\Psi}^{-1}(\mathbf{x}-\boldsymbol{\mu}-\boldsymbol{\Lambda}\mathbf{F}) = \mathbf{0}$, so $\boldsymbol{\Delta}\mathbf{F} = \boldsymbol{\Lambda}^\top\boldsymbol{\Psi}^{-1}(\mathbf{x}-\boldsymbol{\mu})$ and $\hat{\mathbf{F}}^{\text{wls}} = \boldsymbol{\Delta}^{-1}\boldsymbol{\Lambda}^\top\boldsymbol{\Psi}^{-1}(\mathbf{x}-\boldsymbol{\mu})$ (Hessian $2\boldsymbol{\Delta} \succ 0$). Woodbury gives $\boldsymbol{\Lambda}^\top(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi})^{-1} = (\mathbf{I}+\boldsymbol{\Delta})^{-1}\boldsymbol{\Lambda}^\top\boldsymbol{\Psi}^{-1}$, so $\hat{\mathbf{F}}^{\text{reg}} = (\mathbf{I}+\boldsymbol{\Delta})^{-1}\boldsymbol{\Delta}\hat{\mathbf{F}}^{\text{wls}} = (\mathbf{I}+\boldsymbol{\Delta}^{-1})^{-1}\hat{\mathbf{F}}^{\text{wls}}$. With $\mathbf{x}-\boldsymbol{\mu} = \boldsymbol{\Lambda}\mathbf{F}+\mathbf{U}$, $\mathrm{E}[\hat{\mathbf{F}}^{\text{wls}} \mid \mathbf{F}] = \boldsymbol{\Delta}^{-1}\boldsymbol{\Lambda}^\top\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda}\mathbf{F} = \mathbf{F}$, whereas $\mathrm{E}[\hat{\mathbf{F}}^{\text{reg}} \mid \mathbf{F}] = (\mathbf{I}+\boldsymbol{\Delta}^{-1})^{-1}\mathbf{F}$ is shrunk towards 0.

Solution 41 [Ch. 10 — LDA by hand]

(a) $\boldsymbol{\Sigma}^{-1} = \frac{1}{1.75}\begin{pmatrix} 1 & -.5 \\ -.5 & 2 \end{pmatrix}$, $\mathbf{b} = \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1) = (1.429, -1.714)^\top$, midpoint $(2, 1.5)^\top$. $\mathbf{b}^\top(\mathbf{x}_0 - (2, 1.5)^\top) = \mathbf{b}^\top(0, 0.5)^\top = -0.857 < \log(.6/.4) = 0.405$: group 1. Posterior $P(Y = 1 \mid \mathbf{x}_0) = 0.779$.

(b) Let $W = \mathbf{b}^\top(\mathbf{X} - \frac{\boldsymbol{\mu}_1+\boldsymbol{\mu}_2}{2})$. Given $Y = 1$, $EW = \mathbf{b}^\top\frac{\boldsymbol{\mu}_1-\boldsymbol{\mu}_2}{2} = -\frac{\Delta^2}{2}$ and $\mathrm{Var}W = \mathbf{b}^\top\boldsymbol{\Sigma}\mathbf{b} = \Delta^2$. With equal priors an error occurs when $W > 0$: $P(W > 0 \mid Y = 1) = 1 - \Phi(\frac{\Delta^2/2}{\Delta}) = \Phi(-\Delta/2)$; symmetrically for $Y = 2$. Here $\Delta^2 = 32/7 = 4.571$, $\Delta = 2.138$, $\Phi(-1.069) = 0.143$.

(c) With $k = \log(\pi_1/\pi_2) = 0.405$ and rule $W < k \Rightarrow 1$:

$$\pi_1\Phi\Big(\frac{-k - \Delta^2/2}{\Delta}\Big) + \pi_2\Phi\Big(\frac{k - \Delta^2/2}{\Delta}\Big) = 0.6\,\Phi(-1.259) + 0.4\,\Phi(-0.879) = 0.0625 + 0.0758 = 0.138.$$

Solution 42 [Ch. 10 — Fisher's discriminant and QDA]

(a) With $\mathbf{v} = \mathbf{S}_p^{1/2}\mathbf{u}$, $\frac{(\mathbf{u}^\top \mathbf{d})^2}{\mathbf{u}^\top \mathbf{S}_p \mathbf{u}} = \frac{(\mathbf{v}^\top \mathbf{S}_p^{-1/2}\mathbf{d})^2}{\mathbf{v}^\top \mathbf{v}} \le \mathbf{d}^\top \mathbf{S}_p^{-1}\mathbf{d}$ (Cauchy–Schwarz), with equality iff $\mathbf{v} \propto \mathbf{S}_p^{-1/2}\mathbf{d}$, i.e. $\mathbf{u} \propto \mathbf{S}_p^{-1}\mathbf{d}$. The maximum is the squared sample Mahalanobis distance.

(b) $\log f_1 - \log f_2 = -\frac{x^2}{2} + \frac{x^2}{8} + \log 2 = -\frac{3}{8}x^2 + \log 2 > 0 \iff |x| < \sqrt{\frac{8}{3}\log 2} = 1.360$: group 1 inside, group 2 outside. The x^2 terms do not cancel because the variances differ, so the boundary is quadratic (QDA); LDA (equal means here) could not discriminate at all.

Solution 43 [Ch. 10 — generalized Rayleigh quotient]

(a) Put $\mathbf{v} = \mathbf{B}^{1/2}\mathbf{u}$: $\frac{\mathbf{u}^\top \mathbf{A}\mathbf{u}}{\mathbf{u}^\top \mathbf{B}\mathbf{u}} = \frac{\mathbf{v}^\top \mathbf{C}\mathbf{v}}{\mathbf{v}^\top \mathbf{v}}$ with $\mathbf{C} = \mathbf{B}^{-1/2}\mathbf{A}\mathbf{B}^{-1/2}$ symmetric. By the eigendecomposition of $\mathbf{C}$ (Ch. 1), the maximum is $\lambda_1(\mathbf{C})$, attained at its top eigenvector $\mathbf{v}_1$. $\mathbf{C}$ and $\mathbf{B}^{-1}\mathbf{A} = \mathbf{B}^{-1/2}\mathbf{C}\mathbf{B}^{1/2}$ are similar, so $\lambda_1(\mathbf{C}) = \lambda_1(\mathbf{B}^{-1}\mathbf{A})$, and $\mathbf{u}_1 = \mathbf{B}^{-1/2}\mathbf{v}_1$ satisfies $\mathbf{B}^{-1}\mathbf{A}\mathbf{u}_1 = \lambda_1 \mathbf{u}_1$ (the Lagrange condition $\mathbf{A}\mathbf{u} = \lambda \mathbf{B}\mathbf{u}$).

(b) $\mathbf{u}^\top \mathbf{B}\mathbf{u}_j = \mathbf{v}^\top \mathbf{v}_j$, so the constraints are orthogonality to $\mathbf{v}_1, \dots, \mathbf{v}_{k-1}$ in the $\mathbf{v}$-scale; the maximizer is the kth eigenvector $\mathbf{v}_k$ of $\mathbf{C}$, i.e. $\mathbf{u}_k = \mathbf{B}^{-1/2}\mathbf{v}_k$, the kth eigenvector of $\mathbf{B}^{-1}\mathbf{A}$, with value λ_k.

(c) $\mathbf{B}_{\text{MANOVA}} = \sum_\ell n_\ell (\bar{\mathbf{x}}_\ell - \bar{\mathbf{x}})(\bar{\mathbf{x}}_\ell - \bar{\mathbf{x}})^\top$ is a sum of K rank-one terms whose vectors satisfy $\sum_\ell n_\ell(\bar{\mathbf{x}}_\ell - \bar{\mathbf{x}}) = \mathbf{0}$, so its rank is at most $K - 1$ (and at most p). Hence $\mathbf{W}^{-1}\mathbf{B}$ has at most $\min(p, K-1)$ nonzero eigenvalues, i.e. at most that many useful discriminant directions; for $K = 2$ there is one, $\mathbf{W}^{-1}(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)$.

Solution 44 [Ch. 11 — K-means and one EM step]

(a) Initial centroids $(2,2)$ and $(-1,-2)$; $W = (10+10)+(4+4) = 28$. Squared distances to the two centroids: A: 10, 61; B: 10, 9; C: 17, 4; D: 41, 4. B is closer to the CD centroid, so reassign: $\{A\}, \{B, C, D\}$ with centroids $(5,3)$, $(-1,-1)$ and $W = 0 + (4+5+5) = 14$. Recompute: A: 0, 52; B: 40, 4; C: 41, 5; D: 89, 5 — no change. Final clusters $\{A\}, \{B, C, D\}$, $W = 14$.

(b) E-step: $\hat{\delta}_i = \frac{.5\phi(x_i;1,1)}{.5\phi(x_i;0,1)+.5\phi(x_i;1,1)} = \frac{1}{1+e^{-(x_i-0.5)}}$:

$$\hat{\boldsymbol{\delta}} = (0.076,\ 0.182,\ 0.378,\ 0.971,\ 0.989,\ 0.996).$$

M-step: $\pi_1 = \sum \hat{\delta}_i/6 = 0.599$, $\mu_1 = \sum \hat{\delta}_i x_i / \sum \hat{\delta}_i = 4.029$, $\mu_0 = \sum(1-\hat{\delta}_i)x_i / \sum(1-\hat{\delta}_i) = -1.025$. The log-likelihood rises from -36.37 to -13.21.

Solution 45 [Ch. 11 — EM monotonicity and K-means as a limit]

(a) With $\boldsymbol{\delta}$ the missing labels, $p(\mathbf{X}, \boldsymbol{\delta} \mid \boldsymbol{\theta}) = p(\mathbf{X} \mid \boldsymbol{\theta}) p(\boldsymbol{\delta} \mid \mathbf{X}, \boldsymbol{\theta})$, so $\ell(\boldsymbol{\theta}) = \log p(\mathbf{X} \mid \boldsymbol{\theta}) = Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}) - H(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)})$, where $Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}) = \mathrm{E}[\log p(\mathbf{X}, \boldsymbol{\delta} \mid \boldsymbol{\theta}) \mid \mathbf{X}, \boldsymbol{\theta}^{(m)}]$ and $H(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}) = \mathrm{E}[\log p(\boldsymbol{\delta} \mid \mathbf{X}, \boldsymbol{\theta}) \mid \mathbf{X}, \boldsymbol{\theta}^{(m)}]$. By Jensen,

$$H(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}) - H(\boldsymbol{\theta}^{(m)} \mid \boldsymbol{\theta}^{(m)}) = \mathrm{E}\Big[\log \frac{p(\boldsymbol{\delta} \mid \mathbf{X}, \boldsymbol{\theta})}{p(\boldsymbol{\delta} \mid \mathbf{X}, \boldsymbol{\theta}^{(m)})}\Big] \le \log \mathrm{E}\Big[\frac{p(\boldsymbol{\delta} \mid \mathbf{X}, \boldsymbol{\theta})}{p(\boldsymbol{\delta} \mid \mathbf{X}, \boldsymbol{\theta}^{(m)})}\Big] = \log 1 = 0.$$

The M-step gives $Q(\boldsymbol{\theta}^{(m+1)} \mid \boldsymbol{\theta}^{(m)}) \ge Q(\boldsymbol{\theta}^{(m)} \mid \boldsymbol{\theta}^{(m)})$, so $\ell(\boldsymbol{\theta}^{(m+1)}) - \ell(\boldsymbol{\theta}^{(m)}) = \Delta Q - \Delta H \ge 0$.

(b) With equal π_k and common $\sigma^2 \mathbf{I}$,

$$\hat{\delta}_{ik} = \frac{\exp(-\|\mathbf{x}_i - \boldsymbol{\mu}_k\|^2 / 2\sigma^2)}{\sum_j \exp(-\|\mathbf{x}_i - \boldsymbol{\mu}_j\|^2 / 2\sigma^2)} = \frac{1}{\sum_j \exp\{-(\|\mathbf{x}_i - \boldsymbol{\mu}_j\|^2 - \|\mathbf{x}_i - \boldsymbol{\mu}_k\|^2)/2\sigma^2\}}.$$

If $\boldsymbol{\mu}_k$ is the unique nearest mean, every exponent with $j \neq k$ tends to $-\infty$ as $\sigma \to 0$, so $\hat{\delta}_{ik} \to 1$ and $\hat{\delta}_{ij} \to 0$: the E-step becomes the K-means assignment step. The M-step $\boldsymbol{\mu}_k = \sum_i \hat{\delta}_{ik} \mathbf{x}_i / \sum_i \hat{\delta}_{ik}$ then becomes the mean of the points assigned to cluster k: the K-means centroid step.

Solution 46 [Ch. 11 — hierarchical clustering]

The smallest distance $d_{35} = 2$ is merged first by every method.

- **Single**: $d(1,35) = \min(3,11) = 3$, $d(2,35) = 7$, $d(4,35) = 8$, $d_{12} = 9$, $d_{14} = 6$, $d_{24} = 5 \Rightarrow$ (135) at 3. Then $d(2,135) = 7$, $d(4,135) = 6$, $d_{24} = 5 \Rightarrow$ (24) at 5. Finally $d(135,24) = \min(9,6,7,9,10,8) = 6$.
- **Complete**: $d(1,35) = 11$, $d(2,35) = 10$, $d(4,35) = 9$; the minimum is $d_{24} = 5 \Rightarrow$ (24) at 5. Then $d(1,24) = \max(9,6) = 9$, $d(1,35) = 11$, $d(24,35) = \max(7,10,9,8) = 10 \Rightarrow$ (124) at 9. Finally at 11.
- **Average**: (24) at 5 (others: $d(1,35) = 7$, $d(2,35) = d(4,35) = 8.5$). Then $d(1,24) = 7.5$, $d(1,35) = 7$, $d(24,35) = 8.5 \Rightarrow$ (135) at 7. Finally $d(135,24) = \frac{9+6+7+9+10+8}{6} = 8.17$.

Single and average linkage give $\{1,3,5\}, \{2,4\}$ at two clusters; complete linkage gives $\{1,2,4\}, \{3,5\}$: the linkage matters.